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1 Introduction

1.1 Definitions and notation

- **A game is a strategic-form (aka normal-form) game iff all players chose their strategy simultaneously** (and so players chose strategies independently of each other). **Conversely a game is an extensive-form game iff players take turns to chose moves** (and so can alter their approach based on the actions of other players)
- An n-player strategic-form game Γ (capital gamma) is described by: a set $N = [n]$ of players; for each player $i \in N$ their own set S_i of pure strategies (i.e. available moves); for each player $i \in N$ their own payoff function $u_i : S \rightarrow \mathbb{R}$ where $S = \times_{i \in N} S_i$
- Γ is a zero-sum game iff $\forall s \in S; \sum_{i \in N} u_i(s) = 0$

- A strategic-form game Γ is a finite game iff every S_i is a finite set. Let $m_i = |S_i|$ (the number of moves available to player i)
- A mixed strategy $x_i \in X_i$ is a probability distribution over S_i . Thus, $\forall j \in [m_i]; x_i(j) \geq 0$ and $\sum_{j \in [m_i]} x_i(j) = 1$. Let $X = \times_{i \in N} X_i$
- Expected payoff for player i under a given mixed strategy profile $x \in X = U_i(x) = \sum_{s \in S} x(s) u_i(s)$ where $\mathbf{x}(s) = \prod_{j \in [n]} \mathbf{x}_j(s_j) = \mathbf{the\ probability\ of\ } s \mathbf{\ under\ } \mathbf{x}$ (recall players make their choices independently of each other) — note **U_i is the expected payoff of a mixed strategy whereas u_i is the definite payoff of a pure strategy**
- $\pi_{i,j} = \text{the } x_i \in X_i \text{ such that } x_i(j) = 1 \text{ (and so } \forall j' \neq j; x_i(j') = 0) = \mathbf{the\ pure\ strategy\ for\ player\ } i \mathbf{\ in\ which\ they\ play\ move\ } j$

- Given a strategy profile x , let $(x_{-i}; y_i)$ denote the profile that is the same as x for all players except for i who plays strategy y_i instead

1.2 Nash's Theorem

- A strategy profile is a Nash equilibrium iff no player can strictly improve their expected payoff by unilaterally deviating
- $z_i \in X_i$ is a best response for player i to x_{-i} iff
 $\forall y_i \in X_i; U_i(x_{-i}; z_i) \geq U_i(x_{-i}; y_i)$
- Proposition: A strategy profile x is a Nash equilibrium iff
 $\forall i \ x_i$ is a best response to x_{-i}
 Proof: $\forall i \ x_i$ is a best response to $x_{-i} \Leftrightarrow$
 $\forall i \forall y_i \in X_i; U_i(x_{-i}; x_i) \geq U_i(x_{-i}; y_i) \Leftrightarrow$ no player can improve their own payoff by unilaterally deviating from $x \Leftrightarrow x$ is a Nash equilibrium

- **Proposition:** A strategy profile x is a Nash equilibrium $\Leftrightarrow \forall i \in N; \forall j \in S_i; U_i(x) \geq U_i(x_{-i}; \pi_{i,j})$ — **this will be computationally useful** as it tells us that we only need to iterate through pure strategies to verify a candidate (mixed) Nash equilibrium
 Proof: $\Rightarrow$ is trivial, it remains to prove $\Leftarrow$
 Pick an arbitrary player i
 Note that $U_i(x_{-i}; x_i) = \sum_{j \in [m_i]} (x_i(j) U_i(x_{-i}; \pi_{i,j}))$
 Thus, by the hypothesis,

$$U_i(x_{-i}; x_i) \leq \sum_{j \in [m_i]} (x_i(j) U_i(x)) = U_i(x) \sum_{j \in [m_i]} x_i(j) = U_i(x)$$
 Thus, for each player i , x_i is a best response to x_{-i}
 Thus, x is a Nash equilibrium

- Lemma: Let x be a strategy profile. For every player i , there is a move j which they play with strictly positive probability and has pure payoff at most the payoff of x
 Proof sketch: $U_i(x)$ is a weighted average of the payoffs for i of unilaterally deviating from x to each pure strategy weighted by their probability in x_i . It would be absurd for every (non-zero weighted) term in the average to be greater than the value of the average
- Theorem (Brouwer's Fixed Point Theorem, 1909): Every continuous function $f : D \rightarrow D$ mapping a compact and convex non-empty $D \subseteq \mathbb{R}^m$ to itself has a fixed point ($\exists x^* \in D : f(x^*) = x^*$)
 Proof: Out of scope (uses topology not informatics)
- Lemma: X is a compact and convex non-empty subset of $\mathbb{R}^m$ where $m = \sum_{i \in N} m_i$
 Proof: Out of scope

- **Theorem (Nash's Theorem): Every finite game has a Nash equilibrium**

Proof: Let $\psi_{i,j}(x) = \sigma(U_i(x_{-i}; \pi_{i,j}) - U_i(x))$ where σ denotes clipping negative values to 0. Deduce (by the previous useful proposition) that x is a Nash equilibrium iff $\forall i, j; \psi_{i,j}(x) = 0$

Define $f : X \rightarrow X$ by $\forall i, j; f(x)_i(j) = \frac{1}{1 + \sum_{k \in [m_i]} \psi_{i,k}(x)} (x_i(j) + \psi_{i,j}(x))$.

Note that the fraction merely renormalises so that we obtain a new valid probability distribution (sums to 1), it is the other term that determines the behaviour of the new distribution. Deduce that $x \in X \Rightarrow f(x) \in X$ and that f is continuous (it is easy to see will never divide by 0, and everything else is composition of trivially continuous functions). Thus, by Brouwer's Fixed Point Theorem, $\exists x^* \in X : f(x^*) = x^*$. We will show that such a x^* must be a Nash equilibrium (although not relevant to this proof it will from our argument also easily follow that every Nash equilibrium is such a x^*).

Let x^* be an arbitrary fixed point of f . Then,

$\forall i, j; x_i^*(j) = f(x^*)_i(j) = \frac{1}{1 + \sum_{k \in [m_i]} \psi_{i,k}(x^*)} (x_i^*(j) + \psi_{i,j}(x^*))$. By

grade-school algebraic simplification, $x_i^*(j) \sum_{k \in [m_i]} \psi_{i,k}(x^*) = \psi_{i,j}(x^*)$

Recall that we wish to show that $\forall i, j; \psi_{i,j}(x^*) = 0$. Chose an arbitrary player i . By the lemma $\exists k : \psi_{i,k}(x^*) = 0$ and $x_i^*(k) > 0$. Thus,

$x_i^*(k) \sum_{j \in [m_i]} \psi_{i,j}(x^*) = 0$. As $x_i^*(k) > 0$, $\sum_{j \in [m_i]} \psi_{i,j}(x^*) = 0$ and so as $\forall j; \psi_{i,j}(x^*) \geq 0$, $\forall j; \psi_{i,j}(x^*) = 0$ as required.

- Proposition (Indifference Principle): **Let x^* be a Nash equilibrium.**
 $\forall i, j; x_i^*(j) > 0 \Rightarrow U_i(x_{-i}^*; \pi_{i,j}) = U_i(x^*)$. That is that each player would not change their expected utility if they were to unilaterally deviate to any pure strategy they play with non-zero probability (i.e. each pure strategy that they play with non-zero probability is a best response to the Nash equilibrium) — **this will be useful**

Proof: From our proof of Nash's theorem, $\forall i, j; \psi_{i,j}(x^*) = 0$. That is that $U_i(x_{-i}; \pi_{i,j}) - U_i(x) \leq 0$ and so $U_i(x) \geq U_i(x_{-i}; \pi_{i,j})$ as required

- There may or may not be a pure strategy that is a Nash equilibrium, but there will always be a mixed strategy (recall that pure strategies are merely special mixed strategies) that is a Nash equilibrium
- Some games have more than one Nash equilibrium, but every game has at least one Nash equilibrium
- We have only proved the existence of a Nash equilibrium, how to compute it is not yet apparent (recursion with our Brouwer function may diverge instead of converging to a fixed point)

1.3 Alternative notions of optimality

- Social welfare of $x \in X = \sum_i U_i(x)$
- x is Pareto efficient (Pareto optimal) iff
 $\forall x'; \exists k : U_k(x) < U_k(x') \Rightarrow \exists i : U_i(x) > U_i(x')$
- Proposition: “Maximizes social welfare”, “is Pareto optimal”, and “is a Nash equilibrium” are potentially (but not always) distinct

Proof:

Consider the Prisoner’s Dilemma game $[(2, 2), (0, 3)], [(3, 0), (1, 1)]$. The only Nash equilibrium is the $(1, 1)$ payoff but this does not maximize social welfare (note thus cannot possibly be Pareto optimal)

Remaining cases: Exercise

- We call a 2-player game is symmetric iff both players have the same action sets (S_1, S_2) and their payoff matrices are the transposes of each other ($u_1(s_1, s_2) = u_2(s_2, s_1)$) (do not confuse with a zero-sum 2-player game)
- **Proposition:** Every symmetric game has a symmetric Nash equilibrium (a Nash equilibrium in which both players have the same strategy)
Proof: Out of scope
- *Let x_1 be a strategy for (wlog) player 1 in a 2-player symmetric game. x is an evolutionarily stable strategy iff (x_1, x_1) is a (symmetric) Nash equilibrium and at any additional (i.e. other than x_1 itself) best responses x'_1 to x_1 by player 2 player 1 gets strictly higher utility by still playing x_1 rather than joining the deviation to x'_1*
- *Every evolutionarily stable strategy is a Nash equilibrium (however not vice versa)*

- *There exist (2-player symmetric) games for which there does not exist an evolutionarily stable strategy. Moreover, deciding whether a 2-player symmetric game has an evolutionarily stable strategy is in $NP\text{-Hard} \cap coNP\text{-Hard}$*

2 Linear Programming

2.1 Fourier-Motzkin

- Fourier-Motzkin elimination algorithm for solving linear programmes (a wlog objective function is x_0):

```
1 for i in range n to 1 step -1
2     Rearrange each constraint in which  $x_i$  has
      a non-zero coefficient to be in the form
      either  $\alpha \leq x_i$  or  $x_i \leq \beta$  where  $\alpha, \beta$  are linear
      combinations of  $x_0, \dots, x_{i-1}$  (and a constant).
      Call this set of constraints  $H_i$ 
3     Eliminate  $H_i$  from the system of equations
      by removing the constraints in  $H_i$  from our
      working copy of the system and adding new
      constraints  $\alpha_j \leq \beta_l$  for all combinations of  $\alpha$ 
      and  $\beta$  used (if we only used  $\alpha$ s or only
```

- used β s, we do nothing and indeed the variable was redundant). Keep a record however of H_i as we will need to it later
- 4 The single variable system (x_0) that remains can be easily solved (take intersection of intervals of constraints)
 - 5 Substitute the found value of x_0 into H_1 , this gives another single variable system which can be easily solved to obtain x_1 . Then, substitute x_0, x_1 into H_2 . And so on
- Unlike Gaussian elimination (the special case of Fourier-Motzkin elimination where all constraints are equality) which is linear time, Fourier-Motzkin can (in the worst case) square the number of constraints with each variable it removes resulting in an exponential number of increase in the constraints at the end

- Proposition: Every feasible and bounded linear program with rational coefficients has a rational optimal feasible solution (fortunately, imagine having to output an irrational number from a computer!)

Proof: If there is a solution, Fourier-Motzkin will find it, and will do so using additional, multiplication, division of the (rational) coefficients

2.2 Simplex method

- The feasible region of a linear programme is convex, and so a local optimum of the (linear) objective is the global optimum
- To apply the simplex method, rewrite the LP in primal form (maximization objective and all constraints $\leq$) then introduce slack variables to obtain equality constraints

- Call an LP a dictionary iff all constraints (except non-negativity constraints) are equality constraints and have at least one variable which has coefficient 1 and does not appear in any other (equality) constraint nor in the objective function
- Call a set of variables (one from each (equality) constraint) that witnesses an LP being a dictionary a basis
- Example: An LP pre-processed for the simplex method is a dictionary and the slack variables form a basis
- Call a dictionary feasible iff setting its non-basic variables (variables not in its basis) to 0 gives non-negative solutions for the basic variables. Call such a feasible solution to the LP a basic feasible solution

- Simplex method:

```
1 Assume we are given an initial basic feasible
   solution
2 Rearrange so that basis variables are on LHS
   of equality and all other variables are on RHS
3 while True
4     if all variables in objective function
       have non-positive coefficients then we are
       at an optimum and so we should return the
       current basic feasible solution
5     if there is a variable with positive
       coefficient in the objective function and
       non-negative coefficients in all
       constraints then it is unbounded and we
       should return this fact
6     // If we are not done, we pivot
```

- 7 // Bland's rule guarantees no cycling:
Chose the lexicographically smallest (i, j)
(amongst those that obey the below
correctness conditions) (deduce that ties
are much more likely in i than in j). A
rule that is often faster but risks
cycling (but we can always detect cycling
and switch to Bland's) is to pick the x_i
with the largest coefficient in the
objective function
- 8 // Our requirement on i guarantees that
the new bfs will not strictly decrease the
objective, and our requirement on j
guarantees that the new dictionary will be
feasible

- 9 Chose a basic variable x_i with positive coefficient in the objective function to enter the basis and a variable x_j to leave the basis such that the value of x_i in a basic solution to the constraint which x_j is current basic in is minimal.
- 10 Rewrite the constraint with current basic variable x_j to be in the form $x_i = \alpha$
- 11 Substitute α for x_i in all other constraints and the objective function

- We do not need to worry about how we will find an initial basic feasible solution with which to start simplex: For any constraints in which the constant is negative multiply both sides by -1 . Add an extra slack variable to each constraint. Having the extra slack variables be the basis is a trivial bfs for this new LP. Make the objective function of this new LP to minimize the sum of the extra slack variables. Run **simplex starting from our trivial bfs**, if the original LP was feasible we will find an optimal solution with dummy objective value 0 and the bfs at this optimal solution is also a bfs for the original LP (as the extra slack variables are all 0)
- Each simplex pivot takes only polynomial time. However, Simplex can require exponentially many pivots. However, on natural problems Simplex requires linearly many pivots in the number of constraints

- The interior-point method can be used to solve absolutely any LP in polynomial time

2.3 Duality

$$\max cx$$

s.t.

$$(Ax)_i \leq b_i \quad \forall i \in \{1, \dots, d\}$$

$$(Ax)_i = b_i \quad \forall i \in \{d+1, \dots, m\}$$

$$x_1, \dots, x_r \geq 0$$

$$x_{r+1}, \dots, x_n \in \mathbb{R}$$

$\Leftrightarrow$
dual

$$\min b^T y$$

s.t.

$$(A^T y)_j \geq (c^T)_j \quad \forall j \in \{1, \dots, r\}$$

$$(A^T y)_j = (c^T)_j \quad \forall i \in \{r+1, \dots, n\}$$

$$y_1, \dots, y_d \geq 0$$

$$y_{d+1}, \dots, y_m \in \mathbb{R}$$

- **Proposition (Weak Duality):** The dual of a (maximization) LP upper bounds its objective value

Proof: $y^T A \geq c$ by the dual constraints. Thus,

$cx \leq y^T Ax \leq y^T b$ as $Ax \leq b$ by the primal constraints. Thus,
 $cx \leq b^T y$ for all feasible x, y

Corollary: If x^*, y^* are optimal solutions to primal and dual respectively, then $cx^* \leq b^T y^*$

- **Proposition:** Let $-y^*$ be the vector of coefficients of the slack variables in the objective function of the final dictionary used by the simplex algorithm to obtain an optimal solution x^* to a primal LP. Then, y^* is a feasible solution to the dual LP and moreover $cx^* = b^T y^*$

Proof: Out of scope

- **Theorem (Strong Duality):** If both the primal and dual LPs are feasible, then their optimal objective function values **coincide**. If the dual is unbounded, then the primal is infeasible (this follows easily from weak duality). Symmetrically, if the primal is unbounded, then the dual is infeasible. Finally, it is possible that the primal and dual are both infeasible.

Proof sketch: If both are feasible, then simplex will witness the optimal dual solution with the same objective function value as the primal. If one is unbounded, then by weak duality the other must be infeasible

- Proposition (Complementary Slackness): A pair of feasible solutions to primal and dual respectively are optimal iff whenever a primal variable is non-zero the dual constraint is tight (has zero slack) and whenever a dual variable is non-zero the primal constraint is tight

Proof: Recall from our proof of weak duality that $cx \leq y^T Ax \leq y^T b$. By strong duality, $cx^* = (y^*)^T b$. Thus, $cx^* = (y^*)^T Ax$ and so $\left((y^*)^T A - c\right) x^* = 0$

3 (Finite) Strategic Games

3.1 Zero-sum games

- A 2-player game is a zero-sum game iff $\forall s \in S; u_1(s) = -u_2(s)$. Thus a zero-sum game is fully described by a matrix containing only the payoffs for player 1 (row player). Players want to maximize their own payoffs and so player 2 (the column player) wants to minimize the payoffs written in this matrix as their own payoffs or the negatives of these
- Lemma: Let A, B, C be matrices. If $A \geq B$ and $C \geq 0$, then $AC \geq BC$ and $CA \geq CB$
Proof: Out of scope
- Proposition: Let A be a payoff matrix for a 2-player zero-sum game and x_1, x_2 be mixed strategies chosen by player 1 and player 2 respectively. Then, $U_1(x_1, x_2) = x_1^T A x_2$ (and so $U_2(x_1, x_2) = -x_1^T A x_2$)
Proof: Exercise

- A mixed strategy for player 1/2 is a minmaximizer/maxminimizer iff it is a best response by player 1/2 to the worst move for player 1/2's utility that player 2/1 could make
- x_1 is a minmaximizer if $\min_{x_2 \in X_2} (x_1)^T A x_2 = \max_{x_1' \in X_1} \min_{x_2 \in X_2} (x_1')^T A x_2$
- x_2 is a maxminimizer if $\max_{x_1 \in X_1} (x_1)^T A x_2 = \min_{x_2' \in X_2} \max_{x_1 \in X_1} (x_1)^T A x_2'$
- Theorem (Minimax Theorem, von Neumann): **Let A be a matrix defining a 2-player zero-sum game. Then, there exists a unique value v^* such that payoff (to player 1) at minmaximizer = v^* = payoff (to player 1) at maxminimizer.** Moreover, the payoff (to player 1) of every Nash equilibrium is v^* (and so every Nash equilibrium of a 2-player zero-sum game has the same value)
Proof sketch:

It is trivial to see that for any pair of mixed strategies x_1, x_2

$$\min_{x'_2 \in X_2} (x_1)^T A x'_2 \leq (x_1)^T A x_2 \leq \max_{x'_1 \in X_1} (x'_1)^T A x_2.$$

It is also trivial to see that $\max_{x'_1 \in X_1} \min_{x'_2 \in X_2} (x'_1)^T A x'_2 \leq \min_{x'_2 \in X_2} \max_{x'_1 \in X_1} (x'_2)^T A x'_1$

The theorem is that $\max_{x'_1 \in X_1} \min_{x'_2 \in X_2} (x'_1)^T A x'_2 = \min_{x'_2 \in X_2} \max_{x'_1 \in X_1} (x'_2)^T A x'_1$

We will show that every pure strategy of player 2 gives (for every (mixed) strategy of player 1) payoff at least v^* to player 1, and that every pure strategy of player 2 gives (for every (mixed) strategy of player 1) payoff at most v^* to player 1. We will then show that these together are sufficient.

Pick an arbitrary 2-player zero-sum game defined by a matrix A . Pick an arbitrary Nash equilibrium (x_1^*, x_2^*) of the game. Let v^* be the payoff to player 1 of the Nash equilibrium $((x_1^*)^T A x_2^*)$.

As they are strategies in a Nash equilibrium we know that they are best responses to each other: $\forall j; U_1(\pi_{1,j}, x_2^*) \leq U_1(x_1^*, x_2^*)$ and $\forall j; U_2(x_1^*, \pi_{2,j}) \leq U_2(x_1^*, x_2^*)$. By definitions, $U_1(\pi_{1,j}, x_2^*) = (A x_2^*)_j$ and

$U_1(x_1^*, x_2^*) = v^*$, and $U_2(x_1^*, \pi_{1,j}) = ((x_1^*)^T A)_j$ and $U_2(x_1^*, x_2^*) = -v^*$.

Thus we have exactly the inequalities we said would be helpful.

Mixed strategies are weighted averages of the underlying pure strategies, thus our bound on each pure strategy also applies to every mixed strategy (and so even to the (mixed) strategy that maximizes/minimizes the

bounded quantity): $\min_{x_2' \in X_2} (x_1^*)^T A x_2' \geq v^* \geq \max_{x_1' \in X_1} (x_1')^T A x_2^*$. This is exactly

the reverse of our previous trivial inequality, so all the signs must be exact equality: $\min_{x_2' \in X_2} (x_1^*)^T A x_2' = v^* = \max_{x_1' \in X_1} (x_1')^T A x_2^*$.

- **As every (2-player) minimax profile is a Nash equilibrium, the indifference principle applies to them also**

- Unlike general games where computing a Nash equilibrium can be computationally hard, for 2-player zero-sum games computing a minimax profile (and so a Nash equilibrium) can be done in polynomial time using linear programming:

$$\max v$$

$$\text{s.t.}$$

$$(x_1^T A)_k \geq v \quad \forall k \in [m_2]$$

$$\sum_{j \in [m_1]} x_1(j) = 1$$

$$x_1(j) \geq 0 \quad \forall j \in [m_1]$$

- Proposition: minimax theorem $\Leftrightarrow$ strong duality theorem

Proof:

$\Leftarrow$: Exercise

$\Rightarrow$: Encode the primal LP as the symmetric zero-sum game defined by

payoff (block) matrix $M = \begin{bmatrix} 0 & A & -b \\ -A^T & 0 & c \\ b^T & -c^T & 0 \end{bmatrix}$

As this is a symmetric zero-sum game the minimax value is 0 and every minmaximizing strategy is also a maxminimizing strategy. Let $((Y^*, X^*, z), (Y^*, X^*, z))$ be a Nash equilibrium in which $z > 0$ (validity of this restriction is an exercise (assessed assignment)). Thus, by the minimax theorem, $AX^* - bz \leq 0$ and $cz - A^T Y^* \leq 0$ (and $b^T Y^* - c^T X^* \leq 0$). Thus $x^* = \frac{1}{z}X^*$ and $y^* = \frac{1}{z}Y^*$ are feasible solutions to the LP and its dual.

3.2 General games

- We say x_i dominates x_i' iff $\forall x_{-i} \in X_{-i}; U_i(x_{-i}; x_i) \geq U_i(x_{-i}; x_i')$
- We say x_i strictly dominates x_i' iff
 $\forall x_{-i} \in X_{-i}; U_i(x_{-i}; x_i) > U_i(x_{-i}; x_i')$
- Deduce (by the same averaging argument as previously) that to determine (strict) domination it suffices to only compare to pure profiles
- We call a (mixed) strategy a (strictly) dominant strategy iff it (strictly) dominates every other strategy
- Proposition: Every pure strategy played with non-zero probability in a dominant mixed strategy is itself a dominant strategy
Proof sketch: Averaging argument
Corollary: To enumerate all (strictly) dominant strategies it suffices to enumerate all combinations of pure strategies

- Proposition: **If a strategy is strictly dominant, then it is a pure strategy**
Proof sketch: Averaging argument
- We call a (mixed) strategy a strictly dominated strategy iff there exists another strategy that strictly dominates it
- We call a (mixed) strategy a weakly dominated strategy iff there exists another strategy that dominates it and has strictly higher utility than it for at least one player
- To check if a strategy is (strictly) dominated, it does not suffice to only check for dominating pure strategies, it must be compared to mixed strategies also
- A Nash equilibrium will never contain a strictly dominated **strategy**, but all players playing a weakly dominated strategy can be a Nash equilibrium

- **Lemma:** Let A be the payoff matrix for player 1 and B be the payoff matrix for player 2. Assume without loss of generality that all entries in A and B are strictly positive (add a sufficiently large constant). Then, (x_1^*, x_2^*) is a Nash equilibrium iff $\exists y, w_1 : Ax_2^* + y = w_1 \wedge y^T x_1 = 0 \wedge y \geq 0$ and $\exists z, w_2 : Bx_1^* + z = w_2 \wedge z^T x_2 = 0 \wedge z \geq 0$

Proof sketch: w_1, w_2 are the payoffs to player 1 and player 2 at the Nash equilibrium. Every pure strategy played with non-zero probability has payoff exactly w and every pure strategy played with zero probability has payoff at most w . **y, z are the vectors of the slacks between the w and each pure strategy**, these are constrained to be non-negative everywhere and zero when the probability of the pure strategy is non-zero

- By normalizing away w_1, w_2 we can write computing a 2-player Nash equilibrium as solving a so called linear complementarity problem: $Mu + v = 1$ s.t.
 $u^T v = 0 \wedge u > 0 \wedge v \geq 0$ where M is the given block matrix $\begin{bmatrix} 0 & A \\ B & 0 \end{bmatrix}$ and u, v are the vectors $\begin{bmatrix} x'_1 \\ x'_2 \end{bmatrix}$ and $\begin{bmatrix} y' \\ z' \end{bmatrix}$ to be solved for
- Call a basis β a complementary basis iff $\forall k; u_k \notin \beta \vee v_k \notin \beta$
(then $u^T v = 0$ as required)
- Call a basis β an i -almost complementary basis iff
 $\forall k \neq i; u_k \notin \beta \vee v_k \notin \beta$ (then $u^T v = u_i v_i$)

- Lemke-Howson Algorithm:

- 1 Start at the basis $\beta = \{\vec{v}\}$. This is a fully complementary basis but does not obey $u \neq 0$
- 2 Pick arbitrary i and pivot β to a i -almost complementary basis
- 3 while β is not fully complementary
- 4 There exists exactly one j such that u_j, v_j are both not in the basis. Moreover, exactly one of u_j, v_j only just left the basis. Pivot β so that the other one enters the basis.
- 5 β is still an almost complementary basis, and it is one we have not yet considered
- 6 // We have found a fully complementary basis in which $u > 0$ and $v \geq 0$

4 Extensive Games

4.1 Extensive games

- Extensive-form games are typically represented by a game tree
- **Information set = a set of nodes for which a player must chose the same strategy of all nodes within as the player cannot distinguish between them** (due to the presence of hidden information)
- Information not available to a player may be known to another player, or may not be known by any player (chance node)
- **A game is a perfect-information game iff all information sets are singletons**
- A pure strategy for a player in an extensive-form game is a labelling of that player's information sets with actions

- Every player playing a pure strategy only necessarily corresponds to a pure **play (root-to-leaf path in the game tree)** if there are no chance nodes. If there are chance nodes then every player playing a pure strategy corresponds to a probability distribution over plays. Thus, we have to consider expected payoff even if we aren't considering mixed strategies
- Proposition: Every (finite) game can be written in both strategic-form and extensive-form

Proof:

A strategic-form game can be rewritten as an extensive-form game by using information sets to hide the moves of “previous” players from “subsequent” players (i.e. the only role that “time” has in an extensive-form game is that more information is revealed, and it is permitted for information to not be revealed)

An extensive-form game can be rewritten as a strategic-form game by having players commit to a function from nodes to actions rather than committing to actions themselves (note there are exponentially more strategies in the strategic-form rewrite than in the extensive-form)

- A subgame is a subtree of the game tree such that every information set is either entirely in or entirely not in the subtree
- A Nash equilibrium of a game is called a subgame perfect Nash equilibrium iff it is also a Nash equilibrium in every subgame
- Lemma: If a game is a perfect-information, then every subgame is also perfect-information
Proof: Obvious

- **Theorem (Kuhn, 1953):** Every finite perfect-information extensive-form game has a pure strategy (subgame-perfect) Nash equilibrium. A mixed strategy Nash equilibrium is already guaranteed by Nash's theorem, but the perfect-information allows us to strengthen to pure strategy and subgame perfect

Proof (backwards induction algorithm):

```

1 for  $i := 1, \dots, n$ 
2   for each node  $w$  which is  $i$  edges away from
     its nearest leaf
3     if  $w$  is a chance node then calculate
        the expected payoff using the payoffs
        labelled onto the children and label  $w$ 
        with this
4     Otherwise the player that controls  $w$ 
        has in their strategy choosing at  $w$ 
        the action which leads to the child

```

labelled with the highest payoff for.
Label w with the payoffs of that child
for each player

- Kuhn's backwards induction algorithm can be adapted to compute a (not necessarily perfect) subgame perfect Nash equilibrium in a finite extensive-form game with imperfect information (but still perfect recall): Somehow (e.g. convert to strategic-form) solve the bottommost subgame (as now not every subtree is a subgame) then replace it with a leaf labelled with the expected payoff for each player at the subgame Nash equilibrium. Repeat for the next bottommost subgame

4.2 Infinite games

- A game tree can be formalised as a prefix-closed set of words over the alphabet of actions. Then, nodes are strings and the children of a node are those that it is a prefix of. A path is a sequence of nodes where each is the previous one with a single action appended. A complete play is a path that starts at the empty string (root) and for which every non-leaf node (i.e. node for which the set of children is non-empty) in the the path has a child in the path
- *A finite alphabet but unbounded string length allows for infinite horizon games but not infinitely branching games. We will not consider infinitely branching games*
- For an infinite game, payoffs are assigned to complete plays instead of leaves
- It can be more intuitive to represent infinite zero-sum perfect information games as finite graphs rather than infinite trees

- Reachability graph game: Given a finite directed graph. If a designated “goal” vertex is reached, player 1 gets utility +1 and player 2 gets utility −1. If any vertex is reached from which the goal vertex is not reachable (a “deadend”), player 1 gets utility −1 and player 2 gets utility +1.

- Algorithm for reachability game:

```

1  $win_i$  := the set of nodes at which player  $i$  wins
2  $V_i$  := the set of nodes which player  $i$  controls
3  $win'_1$  :=  $win_1$ 
4  $s_1$  :=  $\emptyset$ 
5 repeat
6     for  $v \in V \setminus (win'_1 \cup win_2)$ 
7         if  $v \in V_1 \wedge \exists (v, v') \in E : v' \in win'_1$  then
8              $win'_1 \cup = \{v\}$ 
9              $s_1 \cup = \{v \rightarrow v'\}$ 
10        if  $v \in V_2 \wedge \forall (v, v') \in E : v' \in win'_1$  then
11             $win'_1 \cup = \{v\}$ 

```

12 until win'_1 has not changed

13 Player 1 can win (using strategy s_1) iff $v_0 \in win'_1$

5 Inefficiency of equilibria

5.1 Congestion games

- In a congestion game, players' strategies are subsets of a set of resources and the (dis-)utility of a strategy is the sum of cost functions for each chosen resource which are functions of how many other players chose it
- In a network congestion game, resources are edges in a directed graph and strategies are source-to-sink paths
- **Best response dynamics:** Recall that in any game if every player's strategy is a best response then the strategies are a Nash equilibrium. Consequently, **repeatedly switching the strategy of a player whose current strategy is not a best response to a best response will find a Nash equilibrium if it terminates.** In general, this approach may get stuck in a cycle.

- We call a game a potential game iff
 $\exists \Phi : \forall i \in N; \forall s_{-i} \in S_{-i}; \forall s_i, s'_i \in S_i; cost_i(s_i, s_{-i}) - cost_i(s'_i, s_{-i}) = \Phi(s_i, s_{-i}) - \Phi(s'_i, s_{-i})$ i.e. a single potential function Φ captures the differences in costs of every player
- **Theorem (Rosenthal, 1973): In any potential game, best response dynamics with pure strategies will always terminate**
 Proof sketch: Each step of the best respond dynamics strictly increases the player's utility and so the potential function. At some finite value, the potential function attains a maximum (and this coincides with a pure Nash equilibrium)
 Corollary: Every potential game has a pure Nash equilibrium (and it can be computed)

- Proposition: **Every congestion game is a potential game**

Proof: We will show that $\Phi(s) = \sum_{r \in R} \sum_{j=1}^{\#(r,s)} c_r(j)$ is a potential function where $c_r(j)$ is the cost of resource r when being used by j players and $\#(r, s)$ is the number of players using resource r in strategy s

Pick arbitrary i, s'_i, s_i, s_{-i} . Let $s = (s_i, s_{-i})$ and $s' = (s'_i, s_{-i})$. Then,

$$\Phi(s) - \Phi(s') = \sum_{r \in R} \left(\sum_{j=1}^{\#(r,s)} c_r(j) - \sum_{j=1}^{\#(r,s')} c_r(j) \right)$$

Partition R into: R_1 resources used in neither s nor s' , R_2 resources used in both s and s' , R_3 resources used in s but not in s' , R_4 resources not used in s but used in s' . Then,

$$\Phi(s) - \Phi(s') = \sum_{i=1}^4 \sum_{r \in R_i} \left(\sum_{j=1}^{\#(r,s)} c_r(j) - \sum_{j=1}^{\#(r,s')} c_r(j) \right)$$

Deduce that $\forall r \in (R_1 \cup R_2); \#(r, s) = \#(r, s')$ and that

$\forall r \in R_3; \#(r, s) = \#(r, s') + 1$ and that $\forall r \in R_4; \#(r, s) = \#(r, s') - 1$.

Thus, $\Phi(s) - \Phi(s') = \sum_{r \in R_3} c_r(\#(r, s)) - \sum_{r \in R_4} c_r(\#(r, s'))$.

Finally an algebraic trick, $\Phi(s) - \Phi(s') = \sum_{r \in R_3} c_r(\#(r, s)) + \sum_{r \in R_2} c_r(\#(r, s)) - \sum_{r \in R_4} c_r(\#(r, s')) - \sum_{r \in R_2} c_r(\#(r, s'))$ [as $\forall r \in R_2; \#(r, s) = \#(r, s')$]. Thus, $\Phi(s) - \Phi(s') = cost_i(s) - cost_i(s')$ as required

Corollary: In any congestion game, best response dynamics with pure strategies will always terminate

Corollary: Every congestion game has a pure Nash equilibrium (and it can be computed)

5.2 Inefficiency of equilibria

- (Pure) Price of Anarchy = approximation ratio between maximum achievable social welfare (total (expected) utility across all players) and social welfare at the (pure) Nash equilibrium which is worst (i.e. minimum) for social welfare — deduce that Price of Anarchy ≥ 1
- (Pure) Price of Stability = approximation ratio between maximum achievable social welfare (total (expected) utility across all players) and social welfare at the (pure) Nash equilibrium which is best (i.e. maximum) for social welfare
- Price of Anarchy/Stability for a class of games = that for the worst game in the class (i.e. for the game which maximizes it)

- Theorem (Christodoulou and Koutsoupias 2005): The Price of Anarchy of any congestion game with linear cost functions is at most $\frac{5}{2}$

Proof: Omitted due to time

- **Lemma (the potential method):** Let G be a potential game with potential function Φ . Let SC be the social cost function and PoS be the price of stability. If

$\exists c, d > 0 : \forall s; c \cdot SC(s) \leq \Phi(s) \leq d \cdot SC(s)$, then $PoS(G) \leq \frac{d}{c}$

Proof: As this is an upper bound on a price of stability we only need to demonstrate a sufficient Nash equilibrium rather than argue about all possible Nash equilibria. Let s^* be the strategy profile which minimizes social cost. The strategy profile $\tilde{s}$ which minimizes the potential function must be a (pure) Nash equilibrium. By the hypothesis of the lemma, we have that $c \cdot SC(\tilde{s}) \leq \Phi(\tilde{s})$ and that $\Phi(s^*) \leq d \cdot SC(s^*)$. By the construction of $\tilde{s}$, $\Phi(\tilde{s}) \leq \Phi(s^*)$, and so by transitivity $c \cdot SC(\tilde{s}) \leq d \cdot SC(s^*)$ as required

- Theorem: The price of stability of any congestion game with linear cost functions is at most 2 (this bound is not tight)

Proof sketch: For Rosenthal's potential function $\Phi(s) = \sum_{r \in R} \sum_{j=1}^{\#(r,s)} c_r(j)$, $\forall s; \frac{1}{2}SC(s) \leq \Phi(s) \leq SC(s)$. Thus, by the potential method $PoS \leq \frac{1}{\frac{1}{2}} = 2$

6 Social Choice Mechanisms

6.1 Social choice

- A mechanism f is called truthful iff supplying true preferences is a dominant strategy iff
$$\forall i \in N; \forall >_i; \forall >'_i; \forall >_{-i}; f(>_i, >_{-i}) >_i f(>'_i, >_{-i})$$
- An ordinal social choice function is a function that takes in a vector of preference profiles (ordering relations) and outputs a candidate. A cardinal social choice function is a function that takes in a vector of utility profiles (utility functions) and outputs a candidate
- Cardinal social choice functions are a superset of ordinal social choice functions as an ordinal social choice function corresponds to a cardinal one that only looks at the ordering of utilities rather than their magnitudes also — we will only consider ordinal ones though

- Proposition: If a social choice function is a surjective function, then every candidate has a chance to win (if the preference profile is favourable for it)
Proof: Obvious
- **A social choice function is called Pareto optimal iff**

$$\forall \alpha; (\exists \beta : \forall i; \beta >_i \alpha) \Rightarrow f(>) \neq \alpha$$
- Lemma: **If a social choice function is truthful and surjective, then it is Pareto optimal**
Proof: Omitted due to time
- Proposition: Dictatorship is surjective and truthful
Proof: Each candidate will be elected iff they are the top choice of the dictator voter

- **Theorem (Gibbard-Satterthwaite, 1975):** In the unrestricted domain, when there are $m \geq 3$ candidates, a deterministic social choice function is truthful and surjective (if and) only if it is dictatorial

Proof: Out of scope

Corollary: In the unrestricted domain, when there are $m \geq 3$ candidates, a deterministic social choice function is truthful and Pareto optimal (if and) only if it is dictatorial

- Gibbard-Satterthwaite applies to cardinal rules as well as ordinal rules (because it turns out that every truthful cardinal rule is an encoding of an ordinal rule). However, it does not apply to randomized voting rules (which can be truthful in expectation without encoding a deterministic voting rule)

6.2 Social choice in restricted domains

Omitted due to time (see CS359 for median voting)

7 Auction Mechanisms

7.1 Auctions

- A mechanism with money consists of an allocation function which takes in the valuations given to each item by each agent and gives an allocation of an agent to each item (basically a social choice function) and a payment function which takes in the valuations given to each item by each agent and gives a payment for each agent to make (hopefully related to the allocation, but not necessarily)
- We typically assume that the utility of an agent is their (true) valuation of the allocation minus their payment — this is called quasilinear

- A mechanism is called a Groves mechanism iff the allocation function is to choose an allocation which maximizes social welfare and the payment function for each agent is (a constant determined by the valuations of all the other agents (but not the allocation)) minus the social welfare of the allocation to all other agents
- **Proposition: Every Groves mechanism is truthful**
Proof: Omitted due to time
- A mechanism satisfies voluntary participation iff it is impossible for an agent to receive negative utility
- A mechanism satisfies budget balance iff it is impossible for an agent to be assigned a negative payment to make

- **The Vickrey-Clarke-Groves (VCG) Mechanism is the Groves mechanism where the constant for each participant is the maximum social welfare if they had abstained but everyone else still participated.** Thus the payment they have to make is the difference between the social welfare that would have been achieved by the mechanism had they had not participated and the social welfare (to everyone else) that was achieved by them participating.
- **The VCG mechanism is truthful and satisfies voluntary participation and satisfies budget balance**
Proof: Exercise
- Proposition: A sealed-bid second-price auction is the VCG mechanism with a single item to allocate
Proof: Exercise

7.2 Auctions in restricted domains

- A domain is called a single-parameter domain iff the valuations of an agent i over all outcomes is encoded by a single value v_i which is the valuation by i of each item that they assign non-zero valuation to — this is horrific abuse of notation using v_i both for the valuation function and for the scalar parameter that controls the behaviour of the function but it is the standard notation
- An allocation function f in the single-parameter domain is called monotone iff $\forall i; \forall v_i, v_{-i} : v_i(f(v_i, v_{-i})) \neq 0;$
 $\forall v'_i \geq v_i; u_i(f(v'_i, v_{-i})) \geq v_i(f(v_i, v_{-i}))$
- The critical value of an allocation function in the single-parameter domain for a given agent i and a given counter profile v_{-i} is the largest v_i for which they get 0 value from the outcome (i.e. infinitesimally higher v_i they would get value)

- Theorem (Myerson's characterisation, 1981): A mechanism in which losers pay 0 is truthful on a single-parameter domain iff the allocation function is monotone and the payment of every winner is its critical value

Proof: Omitted due to time

7.3 Revenue maximizing auctions

Omitted due to time